

Exercise set 0 : R intro

Notice, this problem set is intended only for getting to know R, and does not contain any theoretical questions.

Problem 1:

Generate 1000 uniform random numbers in array “x” (Hint command “runif”).

Problem 2:

Make a histogram of 1000 uniform numbers (Hint command “hist”).

Problem 3:

By using vectorized operations, count how many of the 1000 uniform numbers are larger than 0.5. (Hint: see what is returned when typing “x>0.5”).

Problem 4:

Plot the function $\sin(x)$ over the interval $x \in [0, 2\pi]$ (Hint make abscissa using “seq” and do the plotting using “plot”)

Problem 5:

Write a function that solves $a + bx + cx^2 = 0$ for x when a, b, c are input variables.

Problem 6:

Write a function that returns a 95% confidence interval ($= [\bar{x} \pm 1.96\sqrt{s^2/n}]$) for the mean in a normal population when the input is a vector of normal random variables of length n .

Problem 7:

Using a for-loop, write a program that prints the Fibonacci numbers smaller than 100000.